CDATA

Whitepaper

The 25% Accuracy Gap

MCP Provider Performance Across Enterprise Workloads

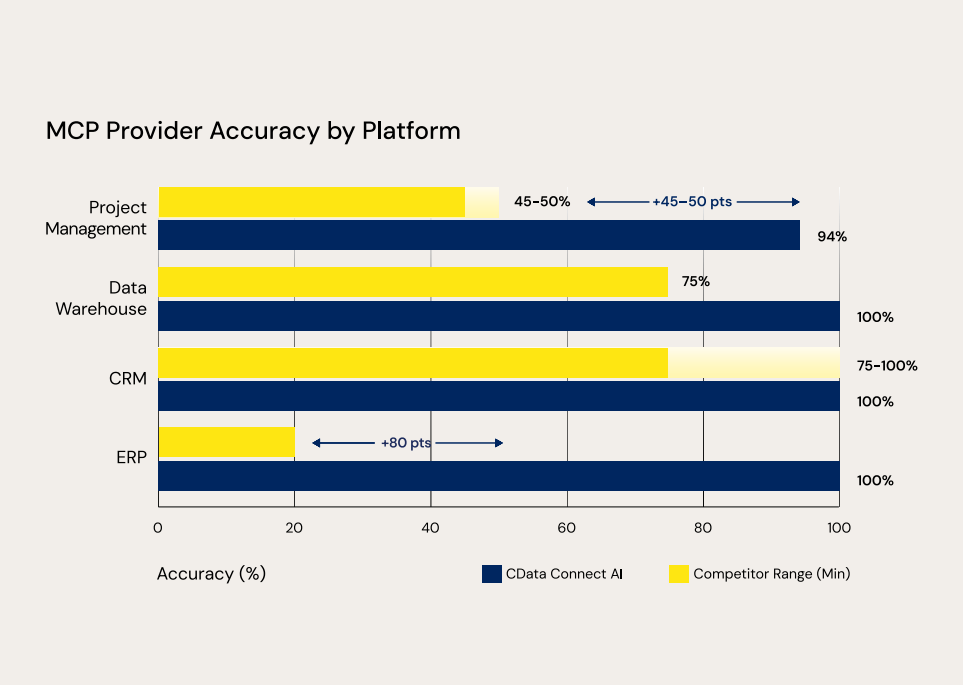
March 2026

Executive Summary

As enterprises deploy AI agents and copilots, they increasingly rely on the Model Context Protocol (MCP) to deliver enterprise context to large language models. But connection alone does not guarantee correctness. In testing across 378 prompts spanning four enterprise platforms (CRM, project management, cloud data warehouse, and ERP), we found that **most MCP servers return incorrect* results 15-42% of the time.**

This report presents our first benchmark of MCP server accuracy for live enterprise application data access. Our findings reveal significant architectural differences, ranging from tool availability to permissioning, that directly impact production AI reliability.

Platform	CData Accuracy	Range for Other Approaches	CData Gap
CRM	100%	75-100%	Consistent
Project Management	94%	45-50%	+45-50 pp
Data Warehouse	100%	75%	+25 pp
ERP	100%	20%	+80 pp
Overall	98.5%	65-75%	+25 pp



* An "all-or-nothing" evaluation was used (a single response was either 0% or 100% correct

Key Findings

- **CData Connect AI achieved 98.5% accuracy (67/68 correct responses), an average of 25+ percentage points higher than other MCP approaches**
- Other MCP approaches showed accuracy rates between 65% and 75%, with errors occurring every 1.5 to 3 queries
- Query complexity directly impacts accuracy: other approaches' accuracy dropped 15- 30 percentage points as queries grew more complex, while CData maintained 98.5% accuracy across all complexity tiers
- Source complexity matters: Project management queries – with their sprint navigation and workflow complexity – averaged only 48.8% accuracy amongst other approaches, while the simple lookup queries for the ERP highlighted CData's architectural advantage – 100% accuracy versus iPaaS's 40% and others' complete failure
- Beyond accuracy, CData achieved an efficiency score of 208.2 (a metric calculated to balance speed and correctness: accuracy ÷ response time × 100), ranking first overall for providers that support multiple platforms.

For enterprises deploying production AI systems where accuracy is non-negotiable, architectural sophistication in MCP servers directly determines reliability. CData Connect AI's source-specific semantic intelligence and standardized relational interface deliver measurably superior accuracy across both information retrieval and action execution – from complex queries and filtering to operations that modify records and trigger state changes in operational systems.

Introduction

The Promise of Model Context Protocol

Large language models have transformed how enterprises interact with data. Instead of writing SQL queries or navigating complex UIs, users can ask questions in natural language and receive actionable insights, while AI agents can take autonomous actions like updating records, triggering workflows, and executing operations across systems. The Model Context Protocol (MCP), introduced by Anthropic, standardizes how AI models access enterprise data sources, enabling consistent, secure connections across CRM systems, data warehouses, project management tools, and ERPs.

But ***connection does not guarantee correctness.*** *As MCP adoption accelerates, we must ask: How accurately do different MCP servers execute real-world enterprise queries?*

The Accuracy Gap

MCP servers take different architectural approaches. Other approaches translate natural language directly to REST API calls. CData's relational interface (a unified query layer with deep semantic intelligence) enables consistent accuracy where direct API translation showed 15-30 percentage point accuracy degradation.

These architectural differences manifest as accuracy gaps in production. A query like "find all deals closing this quarter" requires understanding calendar logic, fiscal year definitions, and CRM-specific field mappings. A request to "move issue PROJECT-12345 to In Progress" requires knowledge of project management workflow validation rules. Get any detail wrong, and the AI returns incorrect data or fails to execute.

How This Benchmark Was Tested

We tested five MCP servers (CData Connect AI, unified API, CRM native, MCP gateway, and iPaaS) across four enterprise platforms: CRM, project management, a cloud data warehouse, and ERP. Each server executed 16 standardized prompts ranging from simple lookups to complex queries with multiple conditions and write operations. This produced 378 total test runs conducted in Q4 2025. An internal evaluator manually assessed each response against pre-established ground truth data.

We measured accuracy (correct vs. incorrect responses), response time, tool calls per query, and token usage. The complete methodology appears in the Methodology section, including query complexity classification, detailed evaluation criteria, and platform-specific test suites.

Queries were classified by complexity: simple queries involved single-entity lookups with basic filtering, medium queries required multiple filter conditions or date calculations, and complex queries involved multi-entity relationships, write operations, or workflow state transitions. This classification enabled us to evaluate how architectural sophistication affects accuracy as task difficulty increases.

What This Benchmark Measures

This report presents internal testing of **five MCP servers** across **four enterprise platforms** (CRM, project management, cloud data warehouse, ERP) executing **378 realworld queries**. We evaluated:

- **Accuracy:** Did the MCP server return correct data or execute the requested action successfully?
- **Performance:** How quickly did queries complete?
- **Token Usage:** How many tokens did each query consume?
- **Efficiency:** What is the composite score of accuracy divided by response time?
- **Consistency:** How does accuracy vary across platforms and query complexity levels?

Why Accuracy Matters for Production AI

In proof-of-concept demos, approximate answers may suffice. In production, **incorrect data erodes trust, leads to poor decisions, and undermines AI adoption.** When a sales executive asks, "which deals are at risk this quarter," an agent that only returns accurate answers 75% of the time actively misleads rather than simply failing to help. When an AI agent automatically updates records using incomplete information, the consequences compound.

Why Accuracy Matters for Production AI

The Real-World Impact of Query Errors

Based on our testing data, here is what accuracy failures look like in practice:

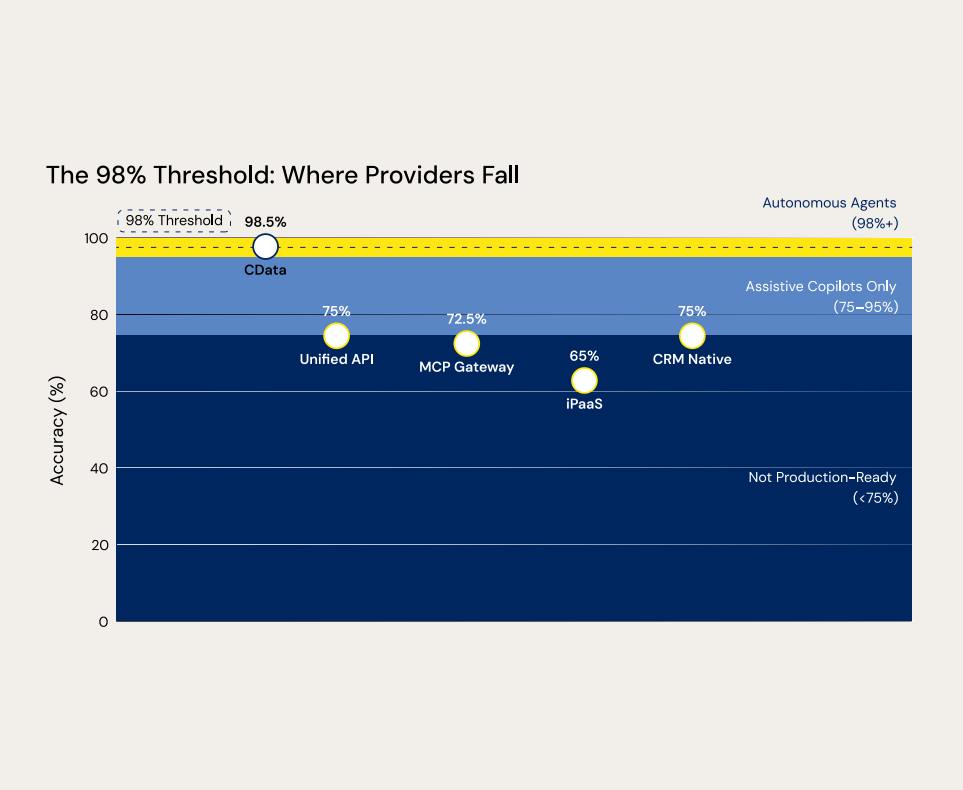
- **Date Logic Errors:** Query: "Find deals closing this quarter." Other approaches returned all deals (missed date filter entirely). Impact: Sales team wastes time reviewing irrelevant pipeline.
- **Filter combination failures:** Query: "Issues assigned to [person] in To Do status." Other approaches returned all issues for that person (ignored status filter). Impact: Project manager gets an incomplete view of workload.
- **Write operation failures:** Query: "Move issue to In Progress." Other approaches attempted updates but failed workflow validation. Impact: Work blocked; manual intervention required.
- **Schema mapping errors:** Query: "Top 10 orders by amount." Other approaches returned the wrong table or column. Impact: Financial decisions based on incorrect aggregations.

Accuracy Requirements by Use Case

In our own internal usage, we have found that different AI applications have different accuracy thresholds based on the level of human oversight:

Use Case	Required Accuracy	Why
Knowledge retrieval (copilots)	High (> 85%)	Human reviews suggestions before acting
Decision support	Very high (> 95%)	Business decisions depend on accuracy
Autonomous agents	**Near perfect (> 98%)**	**No human review before actions execute**

As the industry evolves from assistive copilots to autonomous agents that update CRM records, trigger workflows, and send customer communications without human approval, **accuracy demands increase dramatically.** Production AI agents must deliver consistent performance across diverse scenarios.



Platform and Query Complexity Correlation

Our testing revealed that both platform architecture and query complexity impact MCP server accuracy. Complexity varies based on platform data model and API capabilities. The table below shows average values across all providers.

Platform	Avg Accuracy	Avg Response Time	Avg Tokens Used	Complexity Profile
ERP	46.7%	43.7s	48.778	All simple queries
CRM	91.3%	50.8s	32,400	25% simple, 50% medium, 25% complex
Data Warehouse	87.5%	28.4s	8,700	25% simple, 50% medium, 25% complex
Project Management	**62.5%**	**77.7s**	**34,800**	**50% medium, 50% complex**

These platform-wide averages mask significant variation between MCP servers. The Results section provides detailed accuracy breakdowns, including token usage patterns (not shown above), by server and platform, revealing how architectural differences impact performance across varying complexity levels.

> Key insights
>
> The ERP's overall 46.7% accuracy proves that connectivity maturity determines reliability more than query complexity. The project management platform's 62.5% accuracy reflects the complexity of project management workflows combined with the challenge of executing complex queries that include sprint navigation, priority sorting, and workflow state transitions.

Critically, **CData maintained an average 98.5% accuracy across all platforms and all complexity levels,** while other approaches' accuracy declined 15-30 percentage points as queries grew more complex.

Platform	Connect AI Accuracy	Connect AI Avg. Response Time	Connect AI Avg. Tokens Used	Complexity Profile
ERP	100.0%	44.4s	93,832	All simple queries
CRM	100.0%	42.6s	33,975	25% simple, 50% medium, 25% complex
Data Warehouse	100.0%	36.6s	9,513	25% simple, 50% medium, 25% complex
Project Management	93.8%	59.7s	33,924	**50% medium, 50% complex**

MCP Server Architectures: Why Design Matters

MCP servers differ fundamentally in their design. The architectural approach fundamentally determines accuracy, especially as query complexity increases. This section examines the three primary architectural patterns we observed in testing.

CData Connect AI: Semantic Intelligence

CData Connect AI achieved 98.5% accuracy through a sophisticated three-layer architecture:

- **Source-level Semantic Intelligence:** Rather than translating natural language directly to API calls, CData maintains deep semantic understanding of each source platform. This includes entity relationships ("deals" connect to "contacts" through "associations"), business logic (fiscal calendar definitions, workflow validation rules), and platform-specific conventions (The project management platform's "Highest" priority vs. numeric priority levels).
- **Standardized Relational Interface:** CData translates queries into SQL-like operations, enabling sophisticated filtering, aggregation, and joins across entities. This abstraction handles complex scenarios like "find all deals for Acme Corp with amounts over $50K closing this quarter" with multi-table relationships and date calculations.
- **Mature Enterprise Connector Library:** Decades of connector development ensure CData understands the CRM's deal stages, the project management platform's sprint structure, The ERP's accounting hierarchies, and the cloud data warehouse's schema conventions.

These architectural layers translate into universal MCP tools: the same discovery, read, write, and act operations work identically across all platforms, eliminating the platform-specific inconsistencies that cause failures with other approaches.

The result: CData correctly executed complex queries like "find all issues in Sprint 112 sorted by Highest priority" that require navigating sprint metadata, understanding priority enumeration values, and applying custom sorting logic.

Other Architectures Tested

We tested four additional MCP server implementations representing different architectural approaches:

- **Direct API translation:** Convert natural language to REST API calls with minimal semantic understanding. Fast for simple lookups, prone to errors on multi-step operations.
- **Platform-native MCP:** Vendor-provided MCP servers (e.g., the CRM's official implementation). Strong platform knowledge, limited cross-platform capabilities.
- **Generic connector frameworks:** Unified iPaaS-style connectors adapted for MCP. Broad platform coverage, shallow semantic depth.

Our testing shows that **architectural sophistication directly correlates with accuracy,** particularly on complex queries requiring business logic understanding, multi-entity relationships, or write operation validation.

> "AI technology has advanced faster than organizational data capabilities, creating a critical bottleneck for AI adoption. Ultimately, AI success depends more on having high-quality, well-prepared data than on having the most advanced models."

Harshit Kohli
Sr. Technical Account Manager, AWS

Methodology

This section details our testing approach, including test environment setup, query design, evaluation criteria, and query complexity classification framework.

Test Setup and Environment

LLM Configuration:

- **Model:** GPT-5 (gpt-5)
- **Temperature:** 0.2 (deterministic responses for consistency)
- **Max Tokens:** OpenAI default (not explicitly limited)
- **Agent Pattern:** LangGraph ReAct (Reasoning + Acting) with iterative tool calling
- **Framework:** LangChain with MultiServerMCPClient adapter
- **Testing Period:** Q4 2025 (October-December 2025)
- **Test Data:** Based on real production account structures (not sanitized)

MCP Servers Tested:

- CData Connect AI
- Unified API
- System native MCP
- MCP Gateway
- iPaaS

Platforms Tested:

- CRM
- Project Management
- Data Warehouse
- ERP

Test Scale:

- 378 total test runs
- 16 unique prompts (4 per platform)
- Multiple runs per prompt across different MCP servers

Test Prompts and Design

We tested each platform with four **standardized prompts** that test progressive functionality levels:

CRM Test Suite

- Using my CRM connection, tell me how many deals I have set to close this quarter.
- Using my CRM connection, find contacts named John Smith and show their email and phone number.
- Using my CRM connection, list all deals associated with Acme Corp and include the deal name, amount, and close date.
- Using my CRM connection, change the opportunity Acme Corp Deal to the stage contract sent.

Project Management Test Suite

- Using my project management connection, list all issues assigned to Alex Johnson that are in “To do” status.
- Using my project management connection, list the issues in the ‘Acme’ project that are in the backlog.
- Using my project management connection, find all issues in the Sprint 112 for the Acme project, sorted by Highest priority.
- Using my project management connection, move issue PROJECT-12345 to In Progress.

Data Warehouse Test Suite

- Using my data warehouse connection, show me all tables in the SALESDB schema.
- Using my data warehouse connection, get the top 10 orders by amount from SALESDB.ORDERS.
- Using my data warehouse connection, find all orders created in the last 30 days.
- Using my data warehouse connection, update the status of order 12345 to SHIPPED in SALESDB.ORDERS.

ERP Test Suite

- Using my ERP connection, give me the account numbers and account names for all accounts where account type is Income.
- Using my ERP connection, give me the bin number and last modified date for bins for id starting from 1 to 10.
- Using my ERP connection, list all the charges with id and description in a table.
- Using my ERP connection, give the time zones for countries with id BH and CC.

Evaluation Criteria

A CData Software Triage Engineer 1 manually evaluated all 378 query responses using pre-established ground truth answers. You can review an anonymized version of the test harness in our GitHub:

https://github.com/CDataSoftware/connect-ai-accuracy-test-harness

Correctness Assessment:

We evaluated each response as:

- **Correct:** Response accurately completed the requested task with proper data
- **Incorrect:** Response contained errors, missing data, or failed to execute

Ground Truth Methodology:

Test/demo accounts contain known data sets mirroring real production account structures. Before testing, we established correct answers for each query based on the actual data present in these accounts. This enabled binary evaluation: the MCP server either returned the correct data matching our known ground truth, or it did not.

What constituted 'incorrect':

- Wrong data returned (e.g., incorrect date range, wrong filter applied)
- Incomplete data (missing required fields)
- Failed execution (error messages, timeout, no response)
- Schema/structure errors (wrong table, wrong column names)
- Write operation failures (update not applied, wrong record modified)

Evaluation Limitations:

- Internal evaluators
- No automated validation tools used
- All-or-nothing scoring (partial successes counted as failures)

Query Complexity Framework

To understand where and why MCP servers fail, we classified all 16 test prompts into three complexity tiers based on the technical requirements for accurate execution.

Simple Queries (6 prompts, 40%)

Characteristics:

- Single entity/table access
- Direct ID or type-based lookup
- No date calculations
- Basic filtering only

Examples:

- "Find contacts named John Smith" (CRM)
- "Show all tables in SALESDB schema" (Data Warehouse)
- "List charges with ID and description" (ERP)

Medium Queries (6 prompts, 40%)

Characteristics:

- Multiple filter conditions
- Date-based calculations (relative dates like "this quarter" or "last 30 days")
- Cross-entity relationships
- Aggregation or sorting

Examples:

- "Deals closing this quarter" (requires Q4 2025 date calculation)
- "All deals for Acme Corp with name, amount, close date" (relationship navigation)
- "Top 10 orders by amount" (aggregation + sort)
- "Issues in project that are in backlog" (two filters)

Complex Queries (4 prompts, 20%)

Characteristics:

- Write operations (UPDATE, status transitions)
- Multi-dimensional filtering (3+ conditions)
- Business logic requirements (workflow validation)
- Custom sorting by priority/severity

Examples:

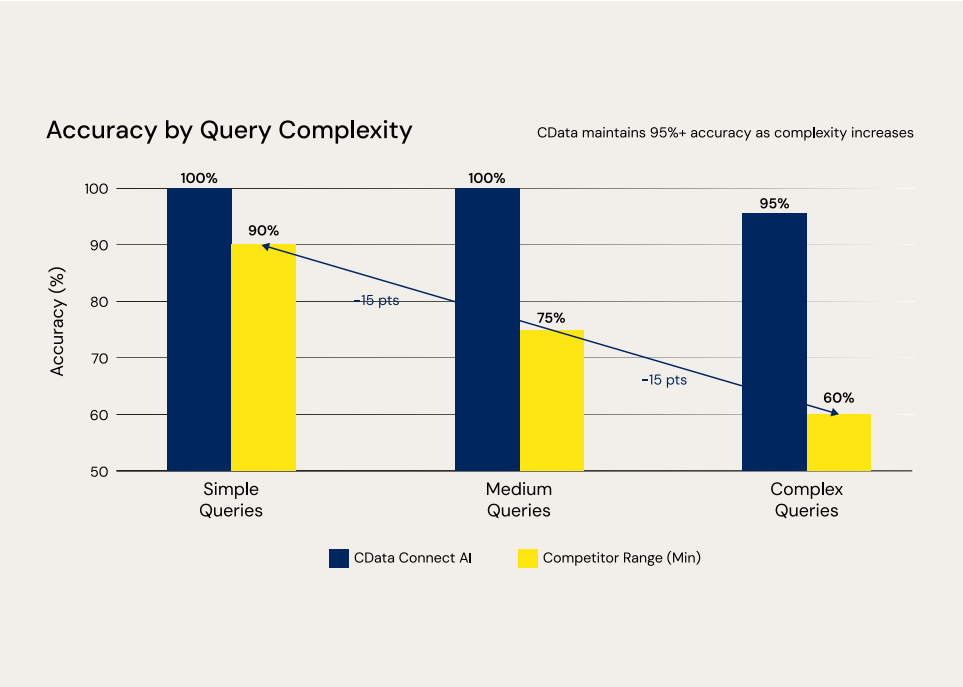
- "Change deal stage to contract sent" (write with validation)
- "Move issue to In Progress" (state transition with workflow rules)
- "Issues in sprint 112 sorted by Highest priority" (sprint filter + priority logic)
- "Update order status to SHIPPED" (write operation)

Complexity Distribution by Platform

The distribution of query complexity varied across platforms based on data model and API capabilities, which helps explain performance differences:

Platform	Simple	Medium	Complex
ERP	4* (100%)	0 (0%)	0 (0%)
Data Warehouse	1 (25%)	2 (50%)	1 (25%)
CRM	1 (25%)	2 (50%)	1 (25%)
Project Management	0 (0%)	2 (50%)	2 (50%)

Hypothesis: MCP servers with weaker semantic intelligence or platform-specific logic would show accuracy degradation as complexity increased. The results section validates this hypothesis.



***NOTE:** Limitations in the ERP's implementation mean that only simple queries were possible.

Performance Metrics

Beyond correctness, we tracked three additional performance dimensions:

- **Response Time:** End-to-end latency from prompt submission to complete response
- **Tool Calls:** Number of MCP tool invocations required to complete the task
- **Token Usage:** Total tokens consumed (input + output) per query

These metrics enabled calculation of an **Efficiency Score:**

$$\text{Efficiency} = (\text{Accuracy Rate} / \text{Avg Response Time}) \times 100$$

This composite metric rewards both speed AND correctness, penalizing fast-but-wrong responses.

Accuracy Results

Overall Accuracy Comparison

We tested five MCP servers across 16 standardized prompts, executing 378 total test runs. The accuracy gap between architectural approaches proved substantial. Run amounts vary based on the provider's support for a given platform.

MCP Server	Correct Responses	Average Accuracy Rate	Gap vs CData
CData Connect AI	67/68	98.5%	—
CRM Native	15/20	75.0%	-23.5 pp
Unified API	30/40	75.0%	-23.5 pp
MCP Gateway	29/40	72.5%	-26.0 pp
iPaaS	47/80	58.75%	-39.75 pp

Key Finding: CData Connect AI maintained 98.5% accuracy while other MCP approaches ranged from 59% to 75%, creating a 23–40-percentage point accuracy gap. This translates to **one error every 68 queries for CData versus one error every 1.5 to 3 queries for other approaches**, depending on query complexity**.**

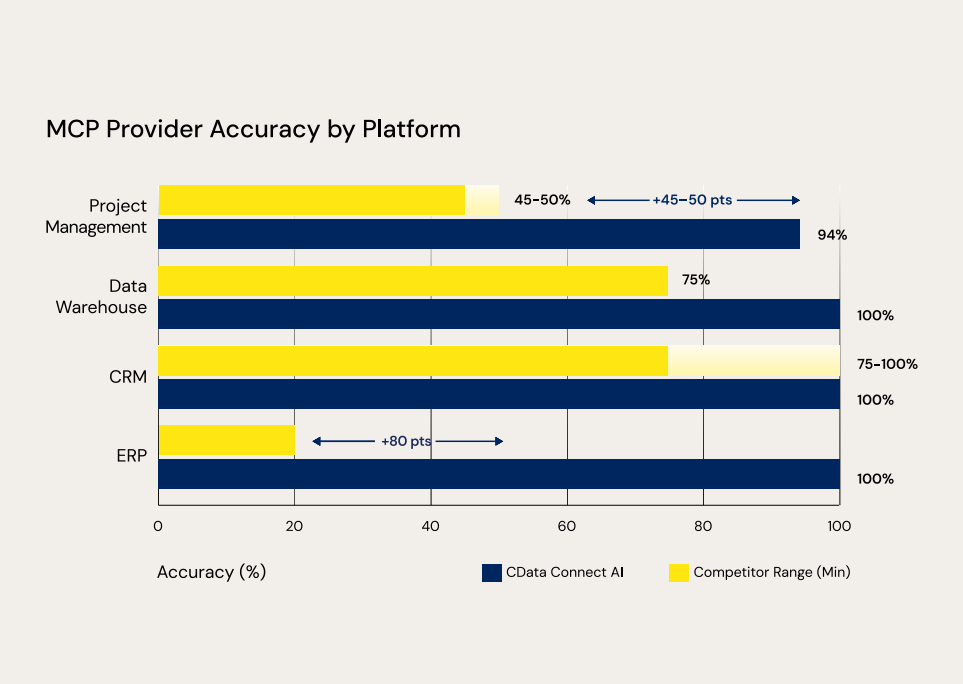
Accuracy by Platform

Performance varied dramatically across platforms, with project management proving most challenging for all servers except CData.

Platform	CData Accuracy	Other Approach Avg	Accuracy Gap
ERP	100%	20%	+80 pp
Project Management	94%	48.8%	+51.2 pp
Data Warehouse	100%	75.0%	+25 pp
CRM	100%	91.3%	+8.7 pp

Platform Insight: The project management platform's 48.8% average accuracy for other approaches reveals the challenge of project management workflow complexity.

Issues requiring sprint navigation, priority-based sorting, and status transitions proved particularly difficult for servers lacking deep platform semantics.



Common Failure Patterns

Analysis of the failures from other approaches revealed recurring architectural weaknesses:

Date Logic Failures (15 occurrences)

Queries like “deals closing this quarter” or “orders created in the last 30 days” require calculating relative date ranges. The iPaaS consistently failed the “this quarter” query, requesting clarification on “specific close date” despite Q4 2025 being unambiguous in context. Multiple servers used wrong date fields (ORDER_DATE instead of CREATED_DATE) or omitted date filters entirely.

Multi-Filter Combination Errors (12 occurrences)

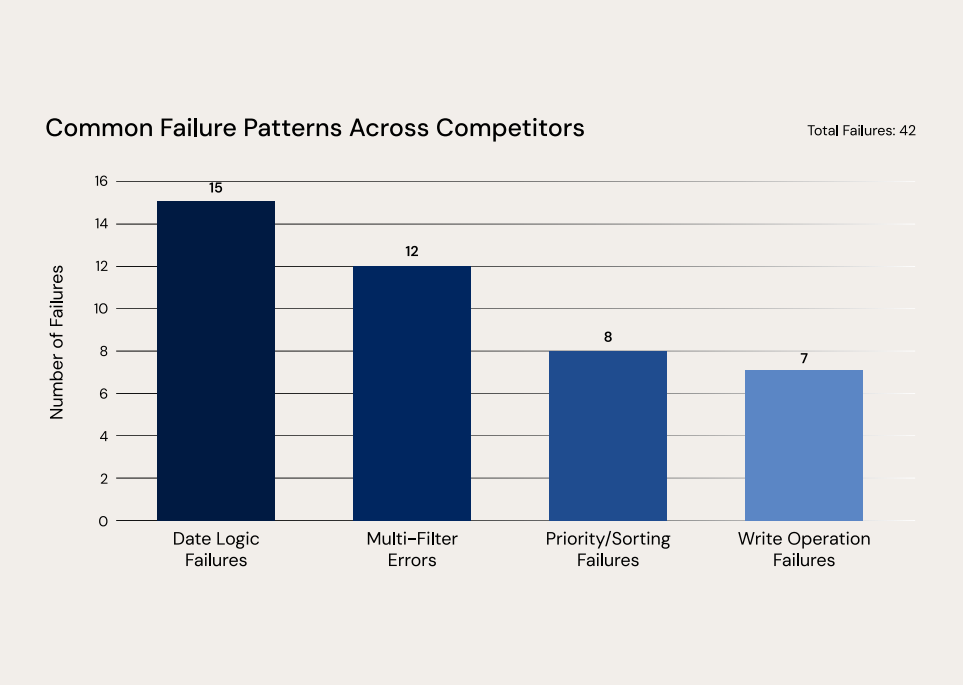
Queries requiring two or more filters proved challenging. “Issues in backlog for project Acme” require both project ID matching AND status filtering. the MCP gateway and the unified API returned all sprint issues (50 tickets) instead of just backlog items (6 tickets), failing to apply the status filter.

Priority/Sorting Logic Failures (8 occurrences)

The “Highest priority” project management query revealed deep semantic understanding gaps. The project management platform uses priority enumeration values (Highest, High, Medium, Low, Lowest) rather than numeric scoring. CData returned “High” priority issues instead of only “Highest” priority. Other approaches either returned all priority levels or failed entirely.

Write Operation Validation Failures (7 occurrences)

Write operations (UPDATE, status transitions) require understanding platform-specific validation rules. The CRM’s native MCP failed all attempts to change the deal stage, due to missing workflow permissions or validation checks. The iPaaS initially failed the project management status transition before succeeding in subsequent runs, suggesting inconsistent validation handling.



Performance & Efficiency Results

Efficiency Score Comparison

To evaluate the **true production value** of MCP servers, we calculated an Efficiency Score combining accuracy and speed:

Efficiency Score = (Accuracy Rate / Avg Response Time) × 100

This composite metric rewards servers that deliver both correct AND fast responses.

MCP Server	Accuracy	Avg Time	Efficiency Score	Rank
CData Connect AI	98.5%	47.3s	208.2	1st
MCP Gateway	72.5%	63.1s	114.9	2nd
Unified API	75.0%	65.6s	114.3	3rd
iPaaS	58.75%	54.3s	108.1	4th

Analysis: CData achieved the highest efficiency score and with 98.5% accuracy, representing the best balance of speed and correctness for production AI systems.

Note: The native MCP servers were not included in this multi-platform efficiency comparison, as they supports only connectivity to a single platform.

The Accuracy-Speed Tradeoff: For autonomous agents that act without human review, accuracy cannot be sacrificed for speed. CData's efficiency score of 208.2 establishes Connect AI as a leader in the combination of speed and accuracy required for production-grade AI agents.

Tool Call Analysis

MCP servers varied in how many tool invocations they required to complete queries:

MCP Server	Avg Tool Calls	Interpretation
CData	6.58	Multi-step reasoning with validation
MCP Gateway	3.69	Moderate complexity
CRM Native	2.81	Streamlined but less thorough
Unified API	1.88	Minimal validation
iPaaS	1.69	Single-step approach

Insight: CData's higher tool call count (6.58 vs. other approaches' 1.69-3.69) reflects architectural sophistication. More tool calls enable semantic validation, error checking, and multi-step query execution—contributing to 98.5% accuracy.

Platform-Specific Insights

CRM Performance

The CRM proved the most mature platform for MCP server implementations, with all servers showing strong performance except on write operations.

CRM Results by Server:

- CData: 20/20 = 100%
- Unified API: 20/20 = 100%
- MCP Gateway: 19/20 = 95%
- CRM native: 15/20 = 75%
- iPaaS: 15/20 = 75%

Average Response Time: 42.6s (CData) vs. 34.1s (CRM native)

Key Finding: Write operations exposed architectural differences. The CRM's native MCP server failed all attempts to update deal stages (0/5 correct), while CData, the unified API, the MCP gateway, and the iPaaS successfully executed deal stage transitions. This suggests that the native implementation lacks workflow validation or permission handling required for write operations.

Why the CRM Was Easiest: Simple query profiles (25% simple, 50% medium, 25% complex) and mature API documentation enabled most servers to perform well on read operations.

Project Management (Project Management) Performance

The project management platform proved the most challenging platform, with accuracy from other approaches averaging just 48.8% compared to CData's 100%.

Project Management Results by Server:

- CData: 15/15 = 100%*
- MCP Gateway: 10/20 = 50%
- Unified API: 10/20 = 50%
- iPaaS: 9/20 = 45%

*CData's only failure across all platforms: returned "High" priority issues instead of "Highest" priority on the sprint query.

Average Response Time: 65.1s (CData) vs. 64.9s-88.4s (other approaches)

Why Project Management Was Hardest:

- No simple queries - Test suite contained 50% medium, 50% complex queries
- Sprint navigation complexity - Identifying issues in specific sprints requires understanding sprint metadata and temporal relationships
- Priority enumerator semantics - The project management platform uses text-based priority levels (Highest, High, Medium, Low, Lowest) that require exact string matching, not numeric approximation
- Workflow state transitions - Moving issues between statuses requires understanding project management workflow validation rules

Common Failures:

- "Issues in backlog": Every other approach tested returned 50 issues (all issues) instead of 6 (backlog only)
- "Highest priority": Other approaches tested either failed or returned all priority levels
- Status transitions: 20% failure rate on moving issues to "In Progress"

Data Warehouse (Data Warehouse) Performance

Data warehouse queries focused on SQL execution accuracy, particularly date logic and aggregation.

Data Warehouse Results by Server:

- CData: 16/16 = 100%
- iPaaS: 15/20 = 75%

Average Response Time: 36.6s (CData) vs. 20.2s (iPaaS)

Key Failure Pattern: The iPaaS consistently used ORDER_DATE instead of CREATED_DATE for the "orders in last 30 days" query, returning 5 results instead of the correct 6. This date field confusion (0/5 correct on this query) revealed shallow schema understanding.

Why CData Succeeded: SQL virtualization layer correctly mapped "created" to the CREATED_DATE column while other approaches relied on keyword matching that selected wrong fields.

ERP Performance

ERP testing revealed the maturity gap between enterprise connector providers and newer MCP implementations.

ERP Results by Server:

- CData: 20/20 = 100%
- ERP native: 0/20 = 0%
- iPaaS: 8/20 = 40%

Average Response Time: 43.0s (CData)

Why ERP Native Failed: The native MCP server returned "Record 'account' was not found" errors when attempting native queries, suggesting API limitations or insufficient schema mapping in the native implementation. CData's decades of ERP connector development enabled seamless handling of accounting hierarchies, bin management, and cross-entity queries.

Why iPaaS Failed: The iPaaS was able to successfully retrieve bins, but was inconsistent across other prompts, often from poorly constructed requests or mis-managed rate limits.

Complexity Note: All ERP queries were classified as "simple" (basic lookups by account type, ID ranges, etc.), yet the native implementation still failed, highlighting that even simple queries require mature connector infrastructure for ERP platforms.

Cross-Platform Consistency

CData's Architectural Advantage:

One of CData's most significant strengths is platform-agnostic consistency. CData maintained 100% accuracy across CRM, data warehouse, and ERP, with its single failure occurring on the project management platform's most semantically complex query.

Inconsistency amongst other approaches:

Other servers showed dramatic variance:

- MCP Gateway: 95% on the CRM, 50% on the project management platform
- Unified API: 100% on the CRM, 50% on the project management platform
- iPaaS: 75% on the CRM, 45% on the project management platform, 75% on the data warehouse, 40% on the ERP

Implication for Enterprises: Organizations deploying AI across multiple platforms need consistent accuracy. CData's unified semantic intelligence layer delivers predictable performance regardless of source system architecture.

Conclusion

This benchmark reveals a clear finding: **MCP server architecture fundamentally determines accuracy for Production AI applications.** As query complexity increases, architectural sophistication separates leaders from laggards. Simple lookups give way to multi-filter operations and workflow state transitions.

CData Connect AI's 98.5% accuracy (67/68 correct responses) demonstrates that semantic intelligence, SQL virtualization, and deep platform knowledge enable production-grade reliability. Other MCP approaches, with accuracy rates between 65% and 75%, show measurable gaps that grow wider as query complexity increases.

For enterprises deploying autonomous AI agents where incorrect actions carry real business consequences, **98%+ accuracy is the minimum requirement.** The architectural patterns that deliver this reliability are now clear: semantic layers that understand business logic, abstraction layers that enable sophisticated querying, and mature connectors drawing on decades of integration experience.

As MCP adoption accelerates and AI evolves beyond assistive copilots into autonomous agents, accuracy will increasingly define competitive advantage. This benchmark establishes both the bar for production AI reliability and a methodology for rigorous, reproducible performance testing.

About CData

CData provides connectivity, context, and governance for AI-to-data interactions within organizations. The platform offers live access and data replication across 350+ sources, with semantic intelligence designed to improve the accuracy of AI workloads. CData supports AI infrastructure for Anthropic, Databricks, Microsoft, Google, Palantir, and more than 10,000 customers.

Disclaimer

This benchmark was conducted internally by CData Software, Inc. ("CData") to provide accuracy data for the MCP ecosystem. All testing, evaluation, and analysis were performed by CData employees. Results have not been independently verified.

Key Limitations:

- Testing was conducted in Q4 2025 using specific configurations described in the Methodology section
- Sample sizes varied across MCP servers based on platform support capabilities
- Binary (all-or-nothing) scoring was used; partial correct responses were counted as failures
- The "Efficiency Score" is a CData-developed metric, not an industry standard
- Test complexity profiles differed by platform
- Results reflect controlled test conditions and may not represent performance in all production environments

Performance Variability: Actual performance may vary significantly based on deployment configuration, data characteristics, network conditions, query patterns, LLM model versions, API changes, and platform-specific factors. The platforms and MCP servers tested may have been updated since testing was conducted.

Not Professional Advice: This report is provided for informational purposes only and does not constitute professional, legal, or technical advice. Organizations should conduct their own independent testing using data and queries representative of their specific use cases before making purchasing or implementation decisions.

CData.com

The leading provider of data access & connectivity solutions

CData Software is a leading provider of data access and connectivity solutions. Our standards-based connectors streamline data access and insulate customers from the complexities of integrating with on-premise or cloud databases, SaaS, APIs, NoSQL, and Big Data.